STAFF REVIEW · H446 212 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Stack (LIFO).

2. The array must be sorted.

3. All preceding values greater than the key.

4. Linear O(n); binary O(log n) for sorted input.

I Do — source worked steps

1. Predict first — Start [4,1,3] and compare 4 with 1: swap.

2. Check the method — Now compare 4 with 3: swap, producing [1,3,4].

3. Check the method — The pass is complete and the array is sorted.

We Do — guided CHECK

1. Check the method
Compare 7 with 3: swap to [2,6,3,7].
The complete first pass ends [2,6,3,7], not yet sorted.

You Do — source mark schemes

1. [8] After inserting key 3: [3,8,6,1] [2]; key 6: [3,6,8,1] [2]; key 1: [1,3,6,8] [2]; each key is inserted into the sorted prefix by shifting larger values right [2].

2. [8] int low=0,high=a.Length-1 [2]; while(low<=high) [1]; mid=low+(high-low)/2 [1]; if(a[mid]==target)return mid [1]; if(a[mid]<target)low=mid+1; else high=mid-1 [2]; return -1 [1]. Input must be sorted.

3. [8] Merge sort O(n log n) worst case [2] versus bubble O(n^2) [2]; merge sort typically needs O(n) extra array space [2]; linear search is suitable for a small unsorted collection where sorting before a single search would cost time [2].

Exit Ticket — source answer

1. False. It requires a sorted search sequence.